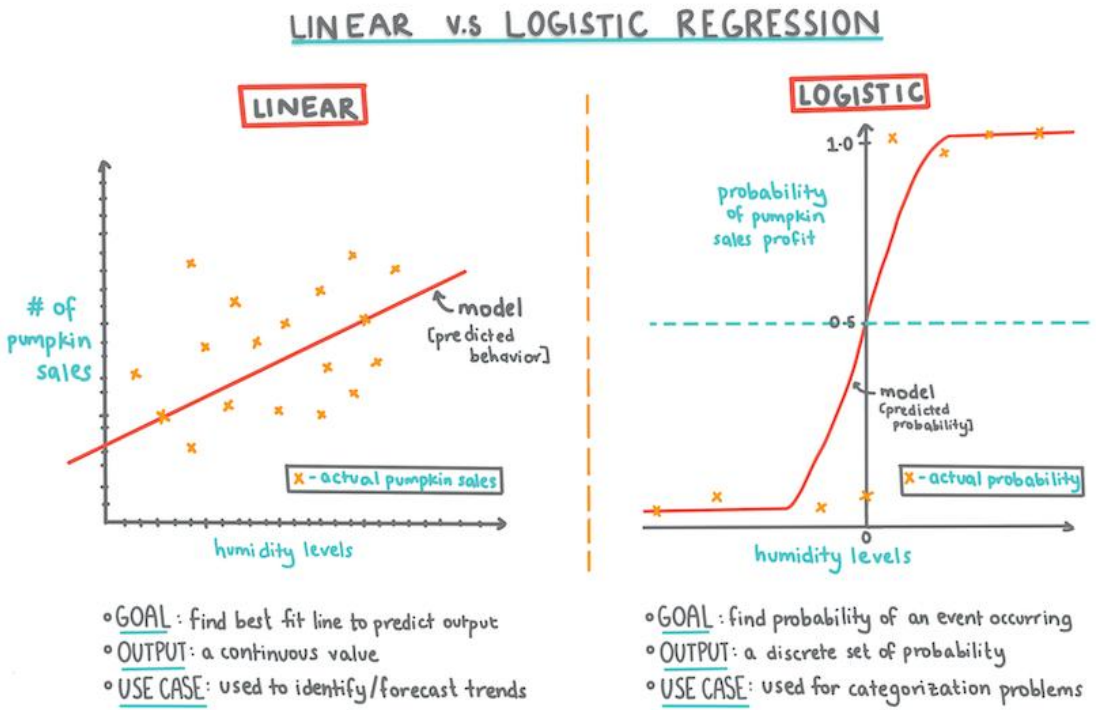
Logistic Regression (LR)



Gerry Tsoukalas

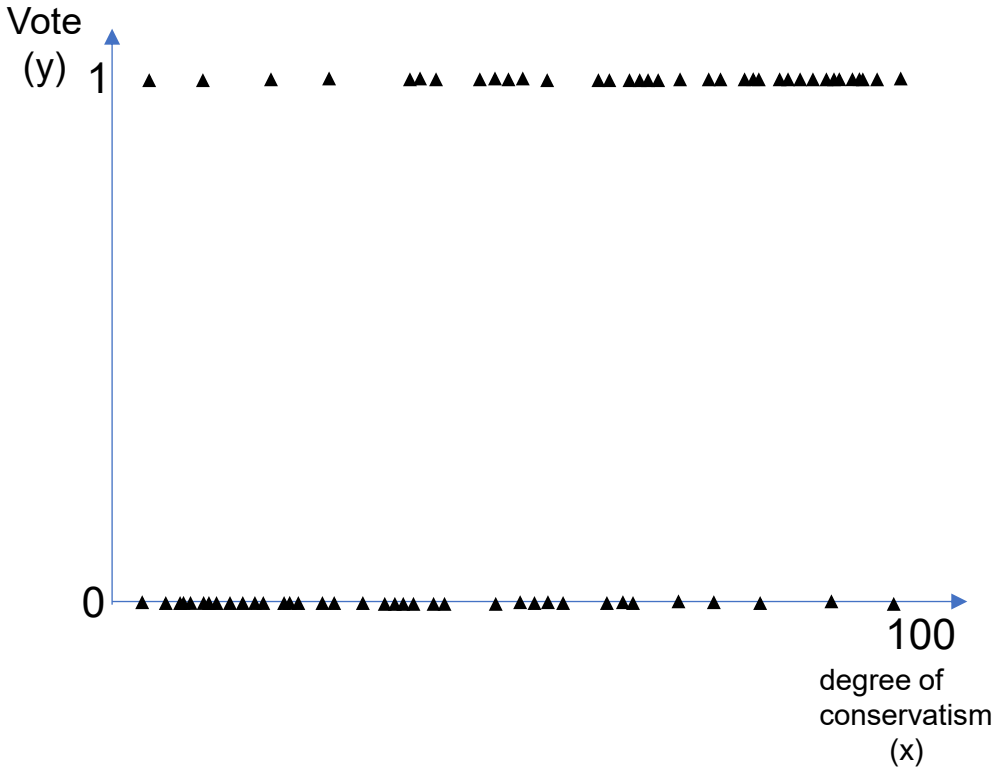
Boston University Questrom School of Business

What is Logistic Regression (LR)?

- An extension of classic “linear regression” to situations where the outcome variable *Y* is categorical, i.e., it is used for classification.
  - Linear regression predicts a metric variable like house prices, stock prices, etc.
  - Logistic regression instead predicts a class
  - Just like in linear reg., LR works better when the predictor variables are (mostly) metric
- Primarily used for binary, $Y=0$ or $Y=1$, classification
- Why isn’t it called logistic classification?
  - It actually predicts a continuous probability, $p(Y=1)$ in [0,1] that is then used for classification
- Where does the word “logistic” come from?
  - Related to the mathematical “logistic function”, aka, sigmoid curve.

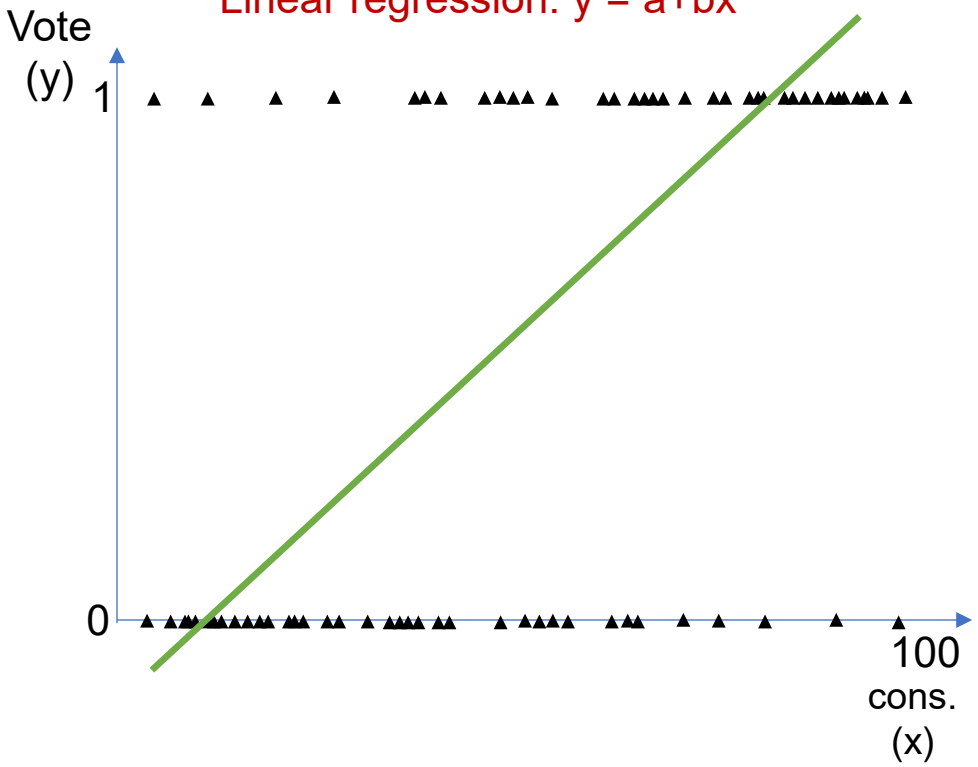
Example: 1999 Clinton Senate Impeachment Vote

- Dependent variable: Vote Not Guilty (0) or Guilty (1)
- Independent/predictor variable: Degree of ideological conservatism
  - 0-100 scale, 100 is most conservative
  - Based on Senator voting records
- Data looks like in the figure…

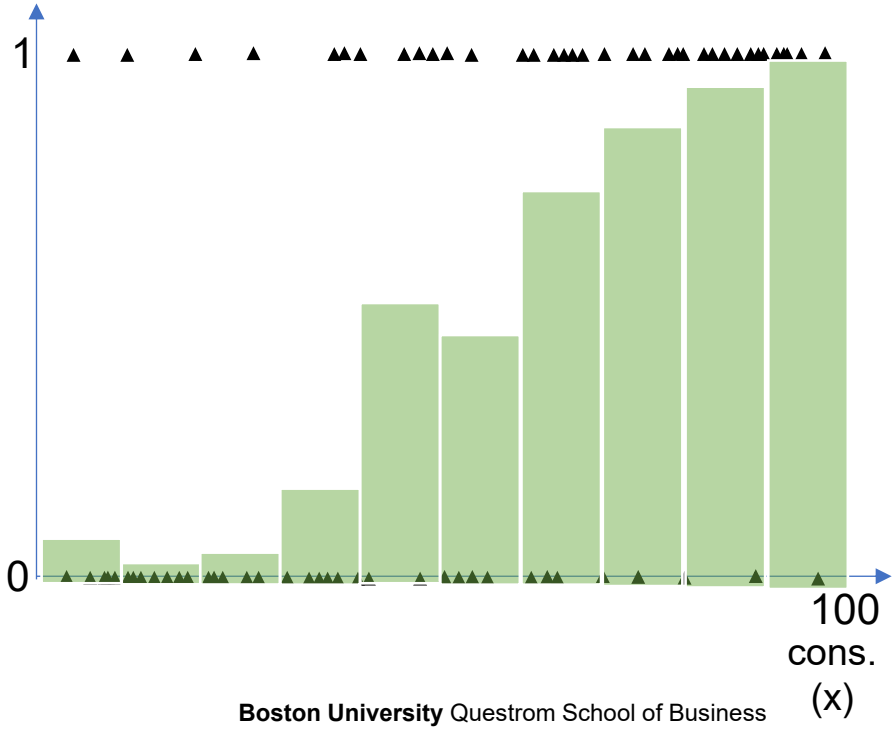


Example: 1999 Clinton Senate Impeachment Vote

Linear regression: y = a+bx

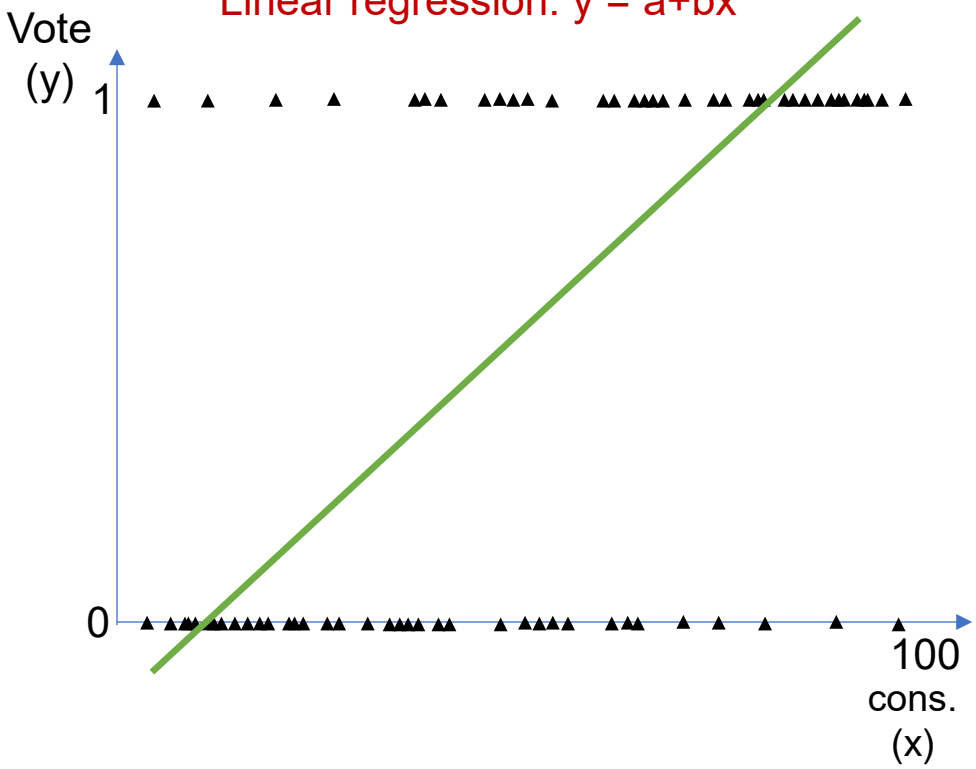


group by “buckets” to get probabilities

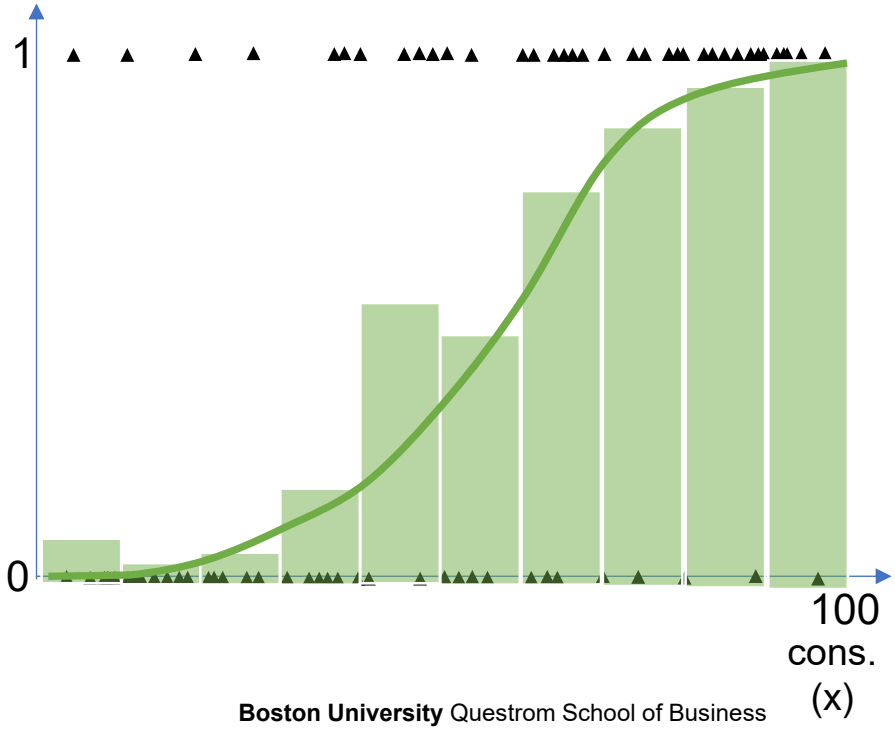


Example: 1999 Clinton Senate Impeachment Vote

Linear regression: y = a+bx



Idea: warp a+bx to fit probabilities

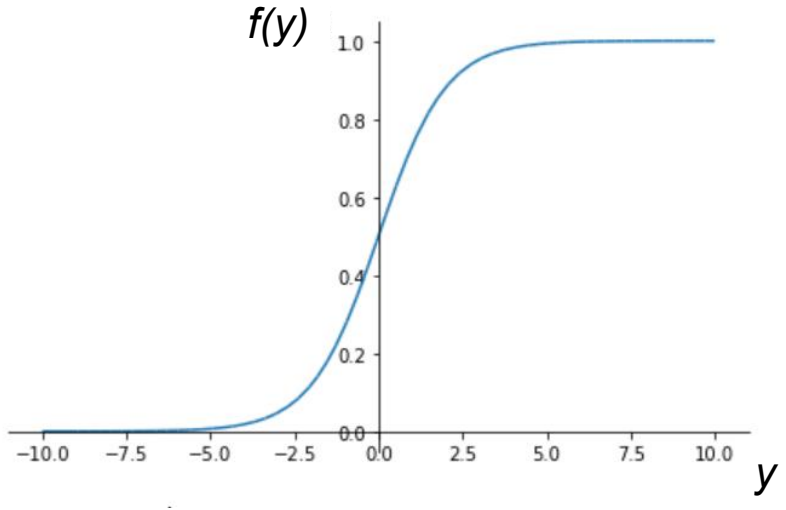


Sigmoid curves / Logistic function

- Basic building block: sigmoid function restricts output *f(y)* to [0,1] for any input *y*

$$f(y) = \frac{1}{1 + e^{-y}}$$

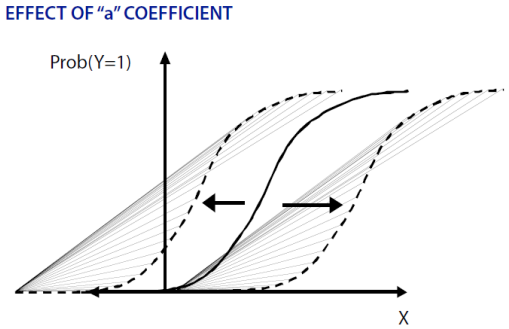
- Useful to model probabilities as it restricts outcomes to [0,1]
- Also a building block for neural networks/deep learning



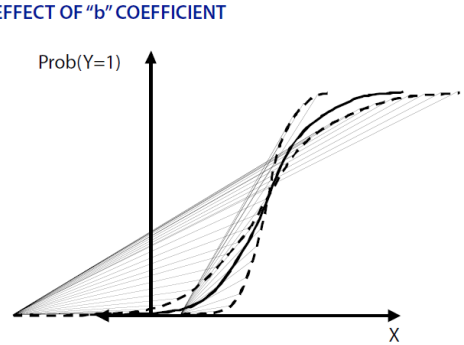
The Logit model: warps linear form a+bx

$$f(x) = \frac{1}{1 + e^{-(a+bx)}}$$

The "a" coefficient shifts the probability curve



The "b" coefficient controls the steepness of the probability curve



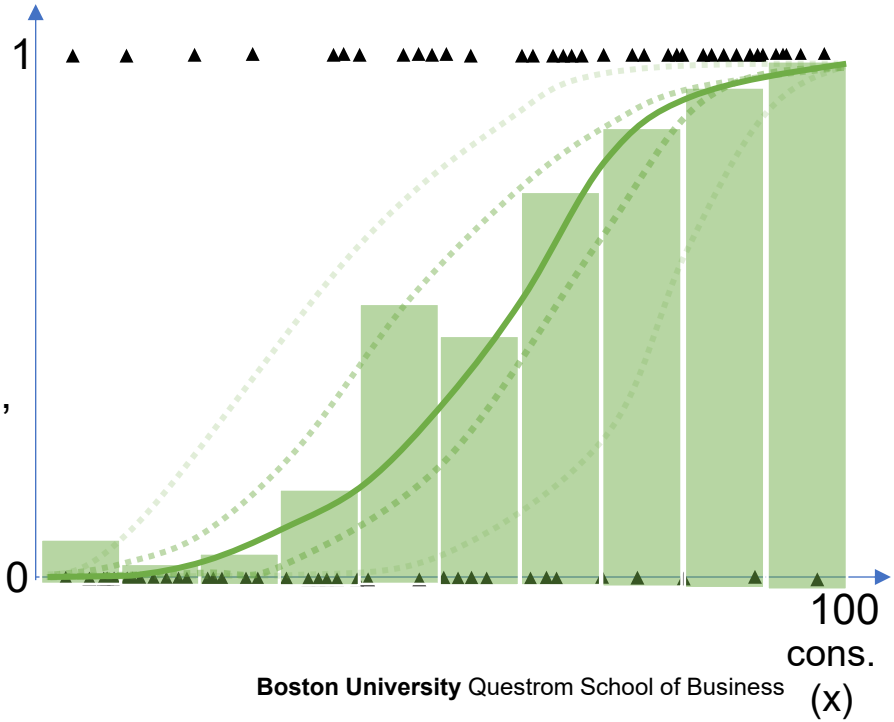
Solving the Logit model

Like in lin reg., we need to search for and find the best parameters *a* and *b* that best fit these probabilities.

In lin. reg., this is done by minimizing RMSE

In log. reg., this is done by a slightly different method to accommodate the fact that we are dealing with binary output 0,1.

The method is called maximizing log-likelihood, and it's done automatically in python



Demo

- Demo: Logistic Regression
- Demo: Sigmoid

How to do Classification?

- Model produces an estimated probability of being a “1”, $p(Y=1)$
- Convert to a classification by establishing cutoff level
- If estimated prob. > cutoff, classify as “1”
- 0.50 is standard initial choice of cutoff, but not necessarily the best one
- Additional considerations for cutoff
  - Maximize classification accuracy
  - Maximize sensitivity (subject to a min. level of specificity)
  - Minimize false positives (subject to a max. false negative rate)
  - Minimize expected cost of misclassification (need to specify costs)

Properties/Definitions of Logistic Regression

- Let p = probability of belonging to class 1, i.e., Prob(Y=1), and 1-p = probability of belonging to class 0, i.e., Prob(Y=0)
- Start with a standard linear function called **logit**: $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots \beta_q x_q$
- Pass through sigmoid to compress to [0,1]: $p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots \beta_q x_q)}}$
- Logit is related to the betting **Odds** (of belonging to class 1): $Odds = \frac{p}{1-p}$
- Relationship between the Odds and the predictors: $Odds = e^{\beta_0 + \beta_1 x_1 + \cdots + \beta_q x_q}$
- Taking the log of Odds, i.e., *log(Odds),* gives us the **logit**: $\log(\text{odds}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_q x_q$
- So, the logit is a linear function of predictors x_1, x_2, …, x_q
- Given the Odds, the probability of belonging to class 1 is: $p = \frac{Odds}{1 + Odds}$

How to Interpret Coefficient Estimates?

- Coefficients are expressed in two forms: original and exponentiated to assist in interpretation
- Directionality of the relationship
  - Original coefficient signs indicate the direction of the relationship
  - Exponentiated coefficients above 1 represent a positive relationship and values less than 1 represent negative relationships
- Magnitude of the relationship
  - Original logistic coefficients are **not useful** in determining the magnitude of the relationship due to the nonlinear nature of the regression
  - But… exponentiated coefficients directly reflect the magnitude of the change in the odds value
    - If we increase x_1 by one unit, to (x_1 +1), holding x_2, x_3 … x_q constant, then e^{β_1} is the factor by which the odds of belonging to class 1 change. Mathematically, this can be seen as follows:

$$New_Odds = e^{\beta_0+\beta_1(x_1+1)+\cdots+\beta_q x_q} = e^{\beta_1} e^{\beta_0+\beta_1 x_1+\cdots+\beta_q x_q} = e^{\beta_1} * \text{Old_Odds}$$

Modeling Considerations (similar to linear reg.)

- Multicollinearity
  - Problem: As in linear regression, if one predictor is a linear combination of other predictor(s), model estimation will not work well
  - Note that in such a case, we have at least one redundant predictor
  - Solution: Remove extreme redundancies (by dropping predictors via variable selection, or by data reduction methods such as PCA)